# Supplementary Materials for Bayesian Optimization of VCSEL Oxide-Aperture Geometry for Polarization-Switching Photonic Ising Machine

https://github.com/Hakainasan/vcsel-polarization-bo

## 1 Question

Photonic Ising machines can use the two orthogonal linear polarization states of a VCSEL as the two values of a spin [1–3]. Real VCSELs carry a built-in anisotropy that pins the light to one polarization (current-driven polarization switching itself has been studied for decades [4, 5]), which *"impedes the system from working as desired"* [2]; tailored oxide apertures can compensate it and produce devices with *"controllable yet easily switchable"* polarization [6]. The question asked here is the one a mask designer faces: **which oxide-aperture geometry gives the best polarization switching for an Ising node, and how should the next fabrication batch be chosen when the constants of the wafer are not yet known?**

## 2 Model

**Mask and aperture geometry.** A cruciform mesa with long-arm length $L$, arm ratio (short/long), arm width $w$, corner radius $\rho w/2$ and rotation $\theta$ of the long arm from the native axis $x$ (the wafer flat, as in [7]). Lateral oxidation removes a depth $d$ from the mesa edge at rates that differ along the two crystal axes (ratio $\eta$): the aperture is the mesa eroded by an ellipse (Minkowski erosion; it rounds concave corners and pinches off narrow arms). Placeholders: $d = 1.0\,\mu$m, $\eta = 1.15$. The boundary is generated analytically and the aperture edge is the zero contour of a smooth erosion field, so every descriptor is continuous in the design parameters.

**Second-rank descriptors.** For a boundary with element normals at angle $\phi$ the shape vector is

$$\mathbf{s} = \oint \omega\,(\cos 2\phi, \sin 2\phi)\,dl \Big/ \oint \omega\,dl.$$

Two weights are used: the intensity of a scalar effective-index fundamental mode ("mode channel"; the solver reproduces the analytic $\mathrm{LP}_{01}$ eigenvalue to 0.11 %) and a long-range kernel $1/(r^2+1)$ ("stress channel"). A $C_4$-symmetric aperture has $\mathbf{s} = 0$ exactly.

**Anisotropy.** Dichroism and birefringence vectors $\mathbf{d} = \mathbf{d}_{nat} + k_d\,\mathbf{s}_{mode}$, $\mathbf{b} = \mathbf{b}_{nat} + k_b\,\mathbf{s}_{stress}$ enter the SFM matrix $M = (d_1 + ib_1)\sigma_z + (d_2 + ib_2)\sigma_x$. Axis convention (as in the experiments): $x$ is the natively favoured axis, $\mathbf{d}_{nat} = -d_0(\cos 2\psi, \sin 2\psi)$ with $d_0 > 0$; $k_d < 0$ means that elongation favours the long axis. The unknown "world" is $(d_0, b_0, k_d, k_b)$ including the *signs* of $b_0, k_d, k_b$. Device-to-device scatter (native $d, b$, axis misorientation 3°, oxidation depth) is integrated with common random numbers.

| **Grounded in the sources read** | |
|---|---|
| Polarization is set by dichroism and birefringence between two linear modes | [8, 9] |
| Tailored oxide apertures compensate native gain anisotropy; the initial polarization follows the elongated arm; a long arm along the native axis strengthens the anisotropy, along the other axis it mitigates it | [6, 7, 10] |
| Switching as the gain sweeps through the polarization resonances; splitting strongly influenced by local strain; splittings from below 1.3 GHz to 9.9 GHz | [4] (used for the prior range $\gamma_p = \pi\,\Delta\nu \approx 4$–$31\,\mathrm{ns}^{-1}$) |
| Lower net anisotropy makes injection flipping about 28 times easier and doubles Ising accuracy in simulation | [2] |
| **Assumptions of this work (not established by any source read)** | |
| Aperture-induced anisotropy is linear in a second-rank boundary tensor; the birefringence channel is an oxide-stress effect with a $1/(r^2+1)$ kernel | assumption |
| Oxidation as erosion by an ellipse with placeholder $d, \eta$; scalar mode used only as a weight; single-mode SFM | assumption |
| Constants $d_0, b_0, k_d, k_b$ and their signs; SFM constants $\kappa = 300, \alpha = 3, \gamma = 1, \gamma_s = 50\,\mathrm{ns}^{-1}$ (typical magnitudes; a second set, [2]: 125, 3, 0.67, 1000, was tried as a robustness check and accepts no world, Sec. 7) | constrained only by the anchors of Sec. 3 |

Table 1: Which parts of the model rest on sources and which are assumptions.

**Yield objective.** A device passes if, sweeping $\mu = I/I_{th}$ over $[1.10, 2.10]$, both branches keep a stable state, exactly one switch occurs at $\mu_{sw} \in [1.25, 1.95]$, and the two states are separated by $\geq 60°$ in azimuth with ellipticity $\leq 0.45$. The hard yield (fraction of devices passing) is reported; the optimiser sees a smoothed version. A device that merely flips polarization at least once in the sweep is said to *switch*; only one that meets the whole specification *passes*, which is rarer (Sec. 8). The window is a specification, not a physical constant: Ising nodes benefit from switching at low bias [10], so a low-bias window is included in the robustness checks.

**SFM solver.** Fixed points (gauge-fixed Newton, analytic Jacobian, global phase removed) and linear stability replace the transient integration of the first version of this project. Against a circular-basis reference right-hand side the residual is $4.5 \times 10^{-13}$. In a sweep over $\gamma_a \in \{1, 2, 3\}\,\mathrm{ns}^{-1}$, $\mu \in [1.1, 2.2]$, $\gamma_p = 20\,\mathrm{ns}^{-1}$, 33 of 36 points agree with time-domain integration; the three disagreements are at $\mu = 2.2$, outside the operating range (`results/sfm_validation.csv`). Results are claimed only for $I/I_{th} \leq 2.1$. The sign convention of $\gamma_a$ (the $x$ mode loses when $\gamma_a > 0$) is the one of Eqs. S1–S2 of [2].

# 3 Anchoring the model to the fabricated devices

The papers report outcomes per design, not numbers. Each outcome is turned into a yes/no test on a simulated device population (24 devices × 3 oxidation-depth nodes) in a candidate world; worlds are drawn from a broad prior (log-uniform magnitudes over one to two decades around the values in the sources, random signs) and kept only if they pass all enforced tests (rejection ABC). Words are turned into thresholds by *choices*: "most" $\geq 0.60$, "seldom" $\leq 0.30$, "part of the devices" $\geq 0.10$ (loose and strict variants: Sec. 7).

| Reported outcome | source | enforced | prior draws passing it alone |
|---|---|---|---|
| A1 circle and square lase along $x$ and (nearly) never switch | [10, 11] | yes | 75 % |
| A2 1:0.7 cruciform with the long arm along $x$ behaves like the circle | [10] | yes | 75 % |
| A3 the initial polarization follows the long arm ($x$ at 0°, $y$ at 90°) | [7, 11] | yes | 29 % |
| A5 the 90° cruciform switches $y \to x$ more often than $x \to y$ | [7] | yes | 10 % |
| A4 the 60° and 90° 1:0.7 cruciforms switch in part of the devices | [10] | yes | 7 % |
| A6 the symmetric cross switches more often than the circle | [11] | no (reported) | 13 % |

Table 2: The anchors. The shares come from a separate run of 300 prior draws in which every anchor is evaluated for every draw; all enforced together: 4.0 % of those draws, against 2.3 % in the main run (181 of 8 000); the two differ by about two standard errors of the smaller run. The 30° member of A4 is reported, not enforced (see below).

**Result.** Of 8 000 prior draws, **181 (2.3 %)** pass every enforced anchor; they are the *literature-consistent worlds* used everywhere below. Median [5th, 95th percentile] of the accepted constants: $d_0 = 1.9$ [1.1, 7.8] ns$^{-1}$, $b_0 = 26.9$ [5.9, 82.4], $k_d = -20.5$ [−51.7, −9.2], $k_b = -28.3$ [−167.9, 36.6]. Three structural facts come out: (i) $k_d < 0$ in every accepted world (an elongated aperture favours its long axis), as A2 and A3 require; (ii) $d_0$ and $|k_d|$ are tightly correlated (Fig. 1): the shape has to be able to cancel the native dichroism at arm ratio 0.7 — a *compensation ridge*, the model's version of the mechanism named in [6]; (iii) 95 % of accepted worlds have $b_0 > 0$ (native birefringence of the class that *cannot* switch, so the circle stays put), while the sign of $k_b$ is not identified (75 % negative). Acceptance depends partly on the device population it was decided on: re-screening the accepted worlds on 3 fresh populations of the same size, 83–92 % still pass all five enforced anchors (nearly all failures are at A4, the anchor with the smallest margin). The fresh-device comparisons of Sec. 6 are therefore slightly outside the set of worlds that the anchors select.

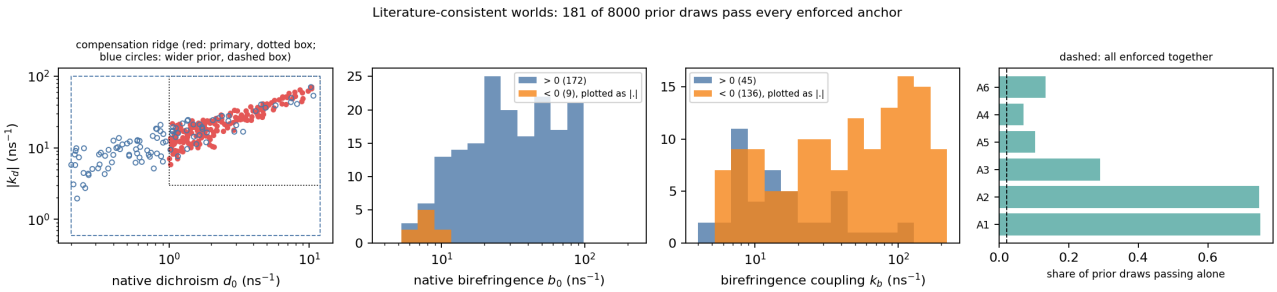



Figure 1: Literature-consistent worlds. Left: native dichroism against dichroism coupling (log axes); red: primary run, dotted: its prior box; blue circles and dashed box: the run with the wider prior. Middle: native birefringence $b_0$ and birefringence coupling $k_b$ (magnitudes, split by sign). Right: share of prior draws that pass each anchor alone (dashed: all enforced anchors together).

**The prior box matters for one constant.** The accepted native dichroism piles up against the lower edge of the prior box ($d_0 \geq 1$ ns$^{-1}$): median 1.95 [1.06, 7.83], and 53 % of the accepted worlds lie below 2 ns$^{-1}$. The anchors do not bound $d_0$ from below; in the model the bound comes from the device-to-device scatter of the native dichroism (0.25 ns$^{-1}$), below which the native polarization is no longer the majority one. A second run with the box widened to $d_0 \geq 0.2$ and $|k_d| \geq 0.6$ ns$^{-1}$ accepted 101 of 4000 draws (2.5 %), with median $d_0 = 0.81$ ns$^{-1}$ and 60 % of the worlds below 1 ns$^{-1}$. What this changes for the ranking of the design sets and for the best rotation is reported in Sec. 7; every other result uses the primary set.

**Two published observations are not reproduced.** (a) The 30° cruciform is grouped with the switching devices in [10]; in the accepted worlds its mean switching fraction is only 0.07 against 0.55 (60°) and 0.58 (90°). (b) The symmetric cross should switch more often than the circle [11]; here 0.02 against 0.00. Both point at

the same structural limit: with a second-rank model a $C_4$ cross carries no shape anisotropy of its own, and at 30°/60° the induced dichroism and birefringence vectors are not parallel to the native ones, which produces elliptical states. Predictions for angles far from 90° should therefore be trusted least; this is a reason to measure them.

# 4 Predicted performance of published and own designs

All numbers of this section are means over the 181 literature-consistent worlds (24 devices × 3 oxidation-depth nodes each); "yield" is the hard yield of Sec. 2. The designs are those of the published experiments [10, 11] and the 16 designs of the author's mask (families A and B, eight rotations; Sec. 8).

| design | mean yield | [10%,90% of worlds] | $P(\text{yield} \geq 0.5)$ | mean $P$(switch) | median $\mu_{sw}$ |
|---|---|---|---|---|---|
| circle | 0.00 | [0.00, 0.00] | 0.00 | 0.00 | 1.78 |
| square | 0.00 | [0.00, 0.00] | 0.00 | 0.01 | 1.25 |
| symmetric cross | 0.00 | [0.00, 0.01] | 0.00 | 0.02 | 1.21 |
| 1:0.7, 0° | 0.00 | [0.00, 0.00] | 0.00 | 0.00 | 1.81 |
| 1:0.7, 30° | 0.02 | [0.00, 0.00] | 0.02 | 0.07 | 1.51 |
| 1:0.7, 60° | 0.30 | [0.00, 0.88] | 0.25 | 0.55 | 1.27 |
| 1:0.7, 90° | 0.43 | [0.03, 0.93] | 0.42 | 0.58 | 1.69 |
| 1:0.6, 0° | 0.00 | [0.00, 0.00] | 0.00 | 0.00 | 1.92 |
| 1:0.6, 90° | 0.06 | [0.00, 0.14] | 0.05 | 0.10 | 1.98 |
| mask A, 0° | 0.00 | [0.00, 0.00] | 0.00 | 0.00 | 1.90 |
| mask A, 30° | 0.03 | [0.00, 0.00] | 0.03 | 0.08 | 1.45 |
| mask A, 45° | 0.09 | [0.00, 0.27] | 0.08 | 0.17 | 1.28 |
| mask A, 60° | 0.32 | [0.00, 0.90] | 0.29 | 0.58 | 1.28 |
| mask A, 90° | 0.43 | [0.03, 0.94] | 0.43 | 0.58 | 1.69 |
| mask A, 120° | 0.32 | [0.00, 0.84] | 0.28 | 0.57 | 1.29 |
| mask A, 135° | 0.09 | [0.00, 0.23] | 0.08 | 0.18 | 1.24 |
| mask A, 150° | 0.03 | [0.00, 0.00] | 0.03 | 0.08 | 1.44 |
| mask B, 0° | 0.00 | [0.00, 0.00] | 0.00 | 0.00 | 1.75 |
| mask B, 30° | 0.01 | [0.00, 0.00] | 0.01 | 0.05 | 1.40 |
| mask B, 45° | 0.06 | [0.00, 0.14] | 0.05 | 0.14 | 1.31 |
| mask B, 60° | 0.25 | [0.00, 0.86] | 0.25 | 0.43 | 1.43 |
| mask B, 90° | 0.13 | [0.00, 0.76] | 0.12 | 0.19 | 1.98 |
| mask B, 120° | 0.24 | [0.00, 0.78] | 0.25 | 0.41 | 1.46 |
| mask B, 135° | 0.06 | [0.00, 0.17] | 0.05 | 0.14 | 1.29 |
| mask B, 150° | 0.01 | [0.00, 0.00] | 0.01 | 0.05 | 1.31 |

Table 3: Predicted yield per design. The bracket gives the 10th and 90th percentile over worlds: the spread between worlds is as large as the differences between designs, so a ranking is a statement about probabilities, not about one wafer.

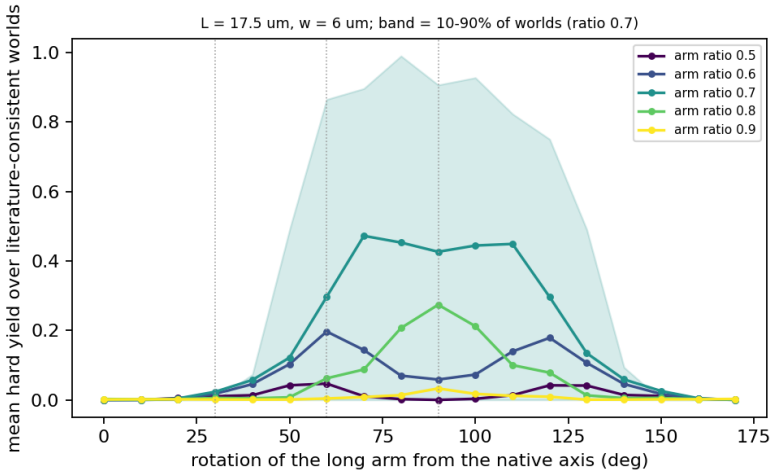



Figure 2: Rotation scan for five arm ratios ($L = 17.5\,\mu$m, $w = 6\,\mu$m, $\rho = 0.5$): mean hard yield over the literature-consistent worlds; the band is the 10–90 % range for arm ratio 0.7.

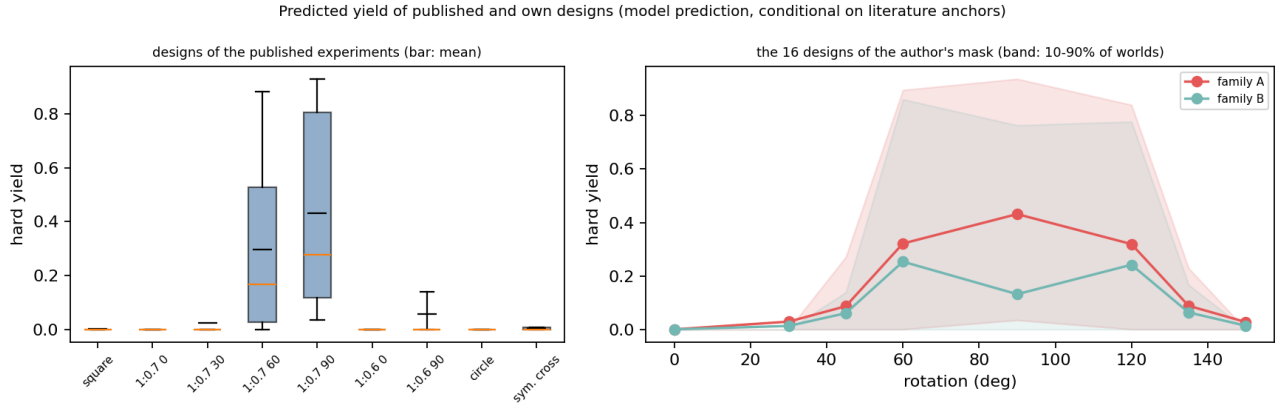



Figure 3: Yield of the designs of the published experiments (left) and of the 16 designs of the author's mask (right).

**Arm ratio and rotation.** For the 1:0.7 cross the mean yield is at least 90 % of its maximum (0.47) for rotations between 70° and 110° (Fig. 2); it is near zero at 0° (long arm along the native axis, no compensation), small at 30° and still on a steep flank at 60°. The share of worlds in which some rotation of the cross (at $L = 17.5$, $w = 6$) reaches a yield of 0.5 depends strongly on the arm ratio: 9 % (0.5), 34 % (0.6), 74 % (0.7), 24 % (0.8), 1 % (0.9). Part of this peak at 0.7 is built into the anchors, which were formulated with the 1:0.7 and 1:0.6 designs of the experiments; it is a property of the *conditional* model, not an independent confirmation that 0.7 is optimal.

| design of APL 126 | share of worlds in which it is the best of the seven |
|---|---|
| square | 0% |
| 1:0.7, 0° | 0% |
| 1:0.7, 30° | 2% |
| 1:0.7, 60° | 31% |
| 1:0.7, 90° | 54% |
| 1:0.6, 0° | 0% |
| 1:0.6, 90° | 4% |

Table 4: Among the seven designs of [10]: the share of worlds in which each is the best one (ties broken toward the smaller index). In 9 % of the worlds none of the seven is a working device.

**Does the ranking of rotations depend on what counts as a good switching current?** Yes, and this matters for the answer below. The yield requires one switch inside a window of $\mu = I/I_{th}$, which is a specification of the node, not a physical constant. Table 5 scores the 1:0.7 cross at rotations from 40° to 90° in 91 worlds (every second literature-consistent world) under four windows.

| window ($\mu_{sw}$) | 40° | 50° | 60° | 70° | 80° | 90° | best | $P(60° > 90°)$ | $P(90° > 60°)$ |
|---|---|---|---|---|---|---|---|---|---|
| default 1.25-1.95 | 0.05 | 0.12 | 0.27 | 0.43 | 0.40 | 0.39 | 70° | 0.31 | 0.63 |
| low bias 1.15-1.60 | 0.05 | 0.11 | 0.27 | 0.39 | 0.27 | 0.23 | 70° | 0.53 | 0.43 |
| wide 1.15-2.05 | 0.07 | 0.16 | 0.36 | 0.54 | 0.48 | 0.46 | 70° | 0.35 | 0.60 |
| mid 1.50-1.70 | 0.01 | 0.04 | 0.06 | 0.12 | 0.13 | 0.12 | 80° | 0.23 | 0.62 |

Table 5: Mean yield of the 1:0.7 cross ($L = 17.5$, $w = 6\,\mu$m, $\rho = 0.5$) at rotations from the native axis, for four switching-current windows (default 1.25–1.95; low bias 1.15–1.60; wide 1.15–2.05; narrow 1.50–1.70), and the share of worlds in which the 60° device beats the 90° one and vice versa (ties excluded).

**Which aperture is best for polarization switching in an Ising node? What the evidence supports.** *Experiments* [10]: the text I read reports injection-locking power for the 1:0.7 cruciform rotated by 60° (248 $\mu$W at 10 mA; 3.6 $\mu$W just above threshold) and for the non-switchable square (371 $\mu$W at 10 mA); it does not rank all seven designs. The 30°, 60° and 90° cruciforms are grouped as switching devices. That is a measurement of injection-locking power on the devices shown, not a yield, and the number of devices per design is not stated in the text I could read. *This model*, conditional on the anchors: among the designs of the published experiments only the 1:0.7 crosses at 60° and 90° have a substantial yield (Table 3); the yield of a 1:0.7 cross rises steeply between 50° and 70°, has a broad plateau between 70° and 110° (the mirror image of 70°–80° is 100°–110°), inside which there is no reliable ranking of the rotations, and falls again beyond it. The best rotation is 70° or 80° in all four windows (Table 5); across the alternative ways of conditioning on the literature of Sec. 7 (Table 18) the plateau is the same but its maximum moves, and 80° (equivalently 100°) is the safest single rotation, though narrowly (Sec. 7). The 90° device is close to the best rotation for the default, wide and narrow windows, but it loses about 40 % of its yield when switching at low bias is required, where the 60° device is as good or better (60° beats 90° in 53 % of the worlds against 43 % for the reverse; for the default window 31 % against 63 %). This is not in conflict with the injection-locking measurement of the 60° device, which was made just above threshold as well as at 10 mA, but it means that "the 60° device was best" and "the 70°–80° device is the most robust choice" are answers to different questions; **the mask of Sec. 8 contains** 60°**,** 90° **and** 120° **and can inform it, but not** 70°–80°. Restricted to the interior of the search box, the first two additions that the model would make to the mask are crosses at 70° and 80° (Table 15); the first unrestricted addition is a thin-armed design at 85° on the wall of the box. *Beyond the published designs*, the batch search of Sec. 6 favours thin-armed crosses with a small arm ratio near 85°–90°; they lie at the edge of the search box, where the reduced model is least reliable (Sec. 6), and are hypotheses, not recommendations. A ranking that is meaningful for the wafer at hand requires measured constants (Sec. 8).

# 5 Optimiser benchmark

**Question.** Does Bayesian optimisation find better apertures than random search, Sobol points and differential evolution for the same number of simulations (48)? **Problems.** *P1*: the wafer constants are known (one of five literature-consistent worlds), default window for the switching current ($\mu_{sw} \in [1.25, 1.95]$). The five wafers were chosen among 30 accepted worlds: those in which no design of a 128-point Sobol cloud reaches a yield of 0.6 were dropped and five spanning the range of difficulty were kept, so they are not a random sample of the accepted worlds. *P2*: as P1 with a narrow window $[1.50, 1.70]$ (good designs are rarer). *P3*: the constants are unknown; one design is chosen to maximise the mean smoothed yield over 8 literature-consistent worlds and is scored on 24 *unseen* worlds — the situation before the first batch is measured. **Methods.** Random search, scrambled Sobol points, differential evolution [12], and Bayesian optimisation with a Gaussian-process surrogate [13] (Matérn-5/2 kernel with one length scale per input, white noise; scikit-learn [14]; the rotation enters as $(\cos 2\theta, \sin 2\theta)$ so that the 180° periodicity is exact), with Expected Improvement [15] or the upper confidence bound [16, 17] maximised over the design box. **Protocol.** The optimisers see the smoothed yield on one fixed device population (common random numbers). The reported quality is the hard yield of the returned design on a *fresh* population of 96

devices (and, for P3, on unseen worlds), so no method benefits from optimism about the sample it saw. Paired Wilcoxon tests and bootstrap intervals are computed over (world, seed) pairs; 20 runs per method for P1 and 8 for P3, 40 for P2.

| | P1 (default window) | P2 (narrow window) | P3 (one design, unseen worlds) |
|---|---|---|---|
| random search | 0.81 [0.74, 0.88] | 0.33 [0.29, 0.37] | 0.39 [0.36, 0.42] |
| Sobol | 0.87 [0.80, 0.92] | 0.35 [0.30, 0.39] | 0.42 [0.38, 0.46] |
| differential evolution | 0.89 [0.84, 0.94] | 0.38 [0.34, 0.42] | 0.37 [0.32, 0.41] |
| GP-EI | 0.96 [0.92, 0.98] | 0.43 [0.38, 0.47] | 0.41 [0.36, 0.45] |
| GP-UCB | 0.86 [0.74, 0.96] | 0.43 [0.38, 0.48] | 0.40 [0.38, 0.43] |

Table 6: Mean held-out hard yield [95 % bootstrap interval] of the design returned by each method.

| problem | paired difference | mean diff. | Wilcoxon $p$ | pairs |
|---|---|---|---|---|
| P1 | GP-EI - random | +0.145 | 0.001 | 20 |
| P1 | GP-EI - Sobol | +0.090 | 0.001 | 20 |
| P1 | GP-EI - DE | +0.063 | 0.008 | 20 |
| P1 | GP-UCB - random | +0.046 | 0.083 | 20 |
| P1 | GP-UCB - Sobol | -0.009 | 0.171 | 20 |
| P1 | GP-UCB - DE | -0.036 | 0.475 | 20 |
| P2 | GP-EI - random | +0.100 | 0.001 | 40 |
| P2 | GP-EI - Sobol | +0.084 | 0.002 | 40 |
| P2 | GP-EI - DE | +0.047 | 0.075 | 40 |
| P2 | GP-UCB - random | +0.100 | 0.001 | 40 |
| P2 | GP-UCB - Sobol | +0.084 | 0.008 | 40 |
| P2 | GP-UCB - DE | +0.047 | 0.139 | 40 |
| P3 | GP-EI - random | +0.018 | 0.641 | 8 |
| P3 | GP-EI - Sobol | -0.014 | 0.844 | 8 |
| P3 | GP-EI - DE | +0.041 | 0.148 | 8 |
| P3 | GP-UCB - random | +0.012 | 0.641 | 8 |
| P3 | GP-UCB - Sobol | -0.020 | 0.547 | 8 |
| P3 | GP-UCB - DE | +0.035 | 0.250 | 8 |

Table 7: Paired differences (method minus baseline) of the held-out yield.

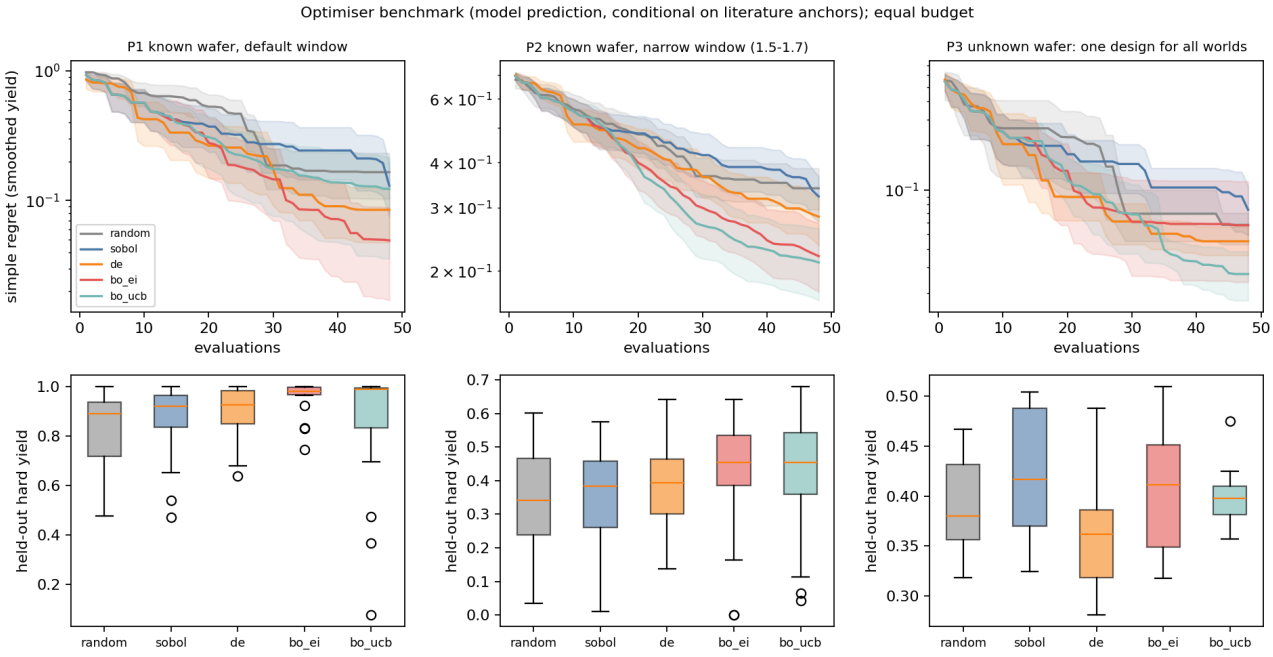



Figure 4: Top: simple regret of the smoothed yield against the best value found by any run in the same world (mean and bootstrap band). Bottom: held-out hard yield.

**Reading of Tables 6 and 7.** *P1 (wafer known, default window).* GP-EI returns designs with a held-out yield of 0.96, against 0.81 for random search, 0.87 for Sobol points and 0.89 for differential evolution; all three differences are significant in paired Wilcoxon tests (+0.14, $p = 0.001$; +0.09, $p = 0.001$; +0.06, $p = 0.008$; 20 paired runs, not corrected for six comparisons, and runs within one world are not independent). At the level of the five wafers (the four seeds of a wafer averaged; Table 8) GP-EI is ahead of random search and of Sobol points on all five wafers (mean difference over wafers +0.14 [+0.06, +0.23] and +0.09 [-0.00, +0.18], 95 % $t$-interval) and of differential evolution on 4 of 5 (+0.06 [-0.03, +0.15]); with only five wafers the interval excludes zero for random search but includes it for Sobol points (marginally) and for differential evolution. GP-UCB is less reliable: its mean (0.86) is not significantly above random search ($p = 0.083$), because 3 of its 20 runs return a design with a yield below 0.5 (GP-EI: 0; random search: 1). In those runs the best value that the optimiser itself had seen on its own device population was also low (0.20–0.49), so they had not converged within 48 evaluations (the fixed exploration weight $\beta = 2$ is the likely reason; I did not test other weights) rather than over-fitted the population. *P2 (wafer known, narrow window).* Both Bayesian optimisers return designs with a held-out yield of 0.43 (GP-EI) and 0.43 (GP-UCB), against 0.33 for random search, 0.35 for Sobol points and 0.38 for differential evolution. The gain over random search (+0.10, $p = 0.001$) and over Sobol points (+0.08, $p = 0.002$) is significant in paired Wilcoxon tests (not corrected for six comparisons; the GP-UCB values are in Table 7); the gain over differential evolution (+0.05, $p = 0.075$ for GP-EI; +0.05, $p = 0.139$ for GP-UCB) is not. By world, the better Bayesian optimiser leads differential evolution by more than 0.02 in 3 of the 5 worlds, is level in 1 and trails in 1; the eight seeds of one world are not independent draws of the problem, so the tests overstate the effective sample size. At the level of wafers (Table 8) GP-EI is ahead of random search on 4 and of Sobol points on 5 of 5 wafers, GP-UCB on 5 and 5; the interval over wafers excludes zero for GP-EI against random search and for GP-UCB against random search and Sobol points, and includes it in every comparison with differential evolution.

| problem | comparison | mean over wafers | 95 % $t$-interval (4 d.f.) | wafers ahead |
|---|---|---|---|---|
| P1 | GP-EI − differential evolution | +0.063 | [-0.028, +0.154] | 4 of 5 |
| P1 | GP-EI − random search | +0.145 | [+0.063, +0.226] | 5 of 5 |
| P1 | GP-EI − Sobol | +0.090 | [-0.001, +0.181] | 5 of 5 |
| P1 | GP-UCB − differential evolution | -0.036 | [-0.269, +0.197] | 3 of 5 |
| P1 | GP-UCB − random search | +0.046 | [-0.088, +0.180] | 3 of 5 |
| P1 | GP-UCB − Sobol | -0.009 | [-0.112, +0.094] | 3 of 5 |
| P2 | GP-EI − differential evolution | +0.047 | [-0.060, +0.154] | 4 of 5 |
| P2 | GP-EI − random search | +0.100 | [+0.024, +0.176] | 4 of 5 |
| P2 | GP-EI − Sobol | +0.084 | [-0.013, +0.181] | 5 of 5 |
| P2 | GP-UCB − differential evolution | +0.047 | [-0.040, +0.133] | 4 of 5 |
| P2 | GP-UCB − random search | +0.100 | [+0.035, +0.166] | 5 of 5 |
| P2 | GP-UCB − Sobol | +0.084 | [+0.004, +0.164] | 5 of 5 |

Table 8: Optimiser benchmark at the level of wafers: the seeds of a wafer are averaged and the five wafer means are compared (mean difference of the held-out yield, 95 % $t$-interval over wafers, number of wafers in which the Bayesian optimiser is ahead).

*P3 (wafer unknown; one design for eight worlds, scored on 24 unseen worlds).* The five methods lie between 0.37 and 0.42 and no difference is significant (Friedman test over the five methods, $p = 0.18$; 8 runs per method, so differences of a few hundredths could not be detected). All five methods lose about the same amount between the eight selection worlds (0.58–0.63, the yield of the returned design on those worlds) and the 24 unseen ones (0.37–0.42; a loss of 0.16–0.24): a design tuned on eight worlds mostly over-fits them, whichever search algorithm produced it, and with eight runs per method the test has little power. This is the situation that motivates the batch of Sec. 6, whose score is a different quantity: the expected best-of-batch yield (0.86 on 32 unseen worlds, 0.74 on 117 further ones) assumes that the best design of the batch can be identified after measurement and is not comparable with the yield of one design. P3 does not show that a single design cannot hedge: the best single design of the pool, chosen on 32 training worlds, reaches 0.69 on 32 other worlds (Table 9), although it lies on a wall of the search box. *What this does not show.* A simulator evaluation costs a fraction of a second, so the budget of 48 is a statement about sample efficiency, not about cost; a fabrication run is many orders of magnitude more expensive and the real batch is 16 designs. The optimiser and the scoring use the same reduced model, so the benchmark says nothing about how well the model represents devices.

# 6 Designing a fabrication batch

A mask is a batch: the 16 designs are fixed before any measurement, and the constants of the wafer are still unknown. The batch is therefore chosen to maximise the *expected best yield in the batch*, $E_w[\max_j Y_j(w)]$ over worlds $w$, a monotone submodular set function that a greedy algorithm maximises within a factor $1 - 1/e$ of the optimum [18]. **Candidates:** 177 designs (128 Sobol points; the best designs of two prior-robust Bayesian-optimisation runs; the nine reference designs; the author's 16; a rotation family). **Selection:** on 32 *training* worlds only. **Scoring:** on 32 disjoint *test* worlds (all worlds are literature-consistent, so the test worlds share the assumptions of the training worlds; they are unseen constants, not an independent check of the model). This is not the batch Bayesian optimisation of [19], which spreads a batch over a Gaussian-process posterior; that alternative was not compared.

| set of designs | $E$[best yield] | $P$(some design $\geq 0.5$) |
|---|---|---|
| greedy portfolio, 16 designs | 0.86 | 0.94 |
| first 8 designs of the portfolio | 0.85 | 0.94 |
| the seven designs of APL 126 | 0.57 | 0.59 |
| author's mask (16 designs) | 0.68 | 0.72 |
| rotation-only family (8, ratio 0.7) | 0.71 | 0.78 |
| 16 random designs from the pool | $0.69 \pm 0.12$ | – |
| best single design (mean yield) | 0.69 (mean) | 0.75 |

Table 9: Expected best-of-batch yield on unseen worlds. In 2 of the 32 test worlds no design of the pool reaches a yield of 0.3.

| | $E$[best yield] | $P$(some design $\geq 0.5$) |
|---|---|---|
| author's mask (16) | 0.68 | 0.72 |
| + 1 added design | 0.84 | 0.88 |
| + 2 added designs | 0.85 | 0.91 |
| + 3 added designs | 0.85 | 0.91 |
| + 4 added designs | 0.86 | 0.91 |
| + 5 added designs | 0.86 | 0.91 |
| + 6 added designs | 0.86 | 0.91 |
| + 7 added designs | 0.87 | 0.94 |
| + 8 added designs | 0.87 | 0.94 |

Table 10: Adding $k$ designs (chosen on the training worlds) to the author's 16-design mask.

| # | $L$ ($\mu$m) | arm ratio | $w$ ($\mu$m) | $\rho$ | rotation (°) |
|---|---|---|---|---|---|
| 1 | 19.6 | 0.50 | 3.5 | 1.00 | 85 |
| 2 | 17.4 | 0.72 | 6.2 | 0.82 | 70 |
| 3 | 17.7 | 0.50 | 3.5 | 0.00 | 90 |
| 4 | 18.6 | 0.78 | 6.3 | 0.52 | 80 |
| 5 | 13.0 | 0.84 | 8.0 | 0.21 | 88 |
| 6 | 17.5 | 0.64 | 4.4 | 0.94 | 36 |
| 7 | 19.3 | 0.67 | 3.8 | 0.16 | 83 |
| 8 | 17.5 | 0.70 | 6.0 | 0.50 | 112 |

Table 11: The added designs of Table 10, in the order of the greedy choice.

| id | origin | $L$ | ratio | $w$ | $\rho$ | $\theta$ | $E$(yield) | $P(\geq 0.5)$ |
|---|---|---|---|---|---|---|---|---|
| 0 | bo1 | 19.7 | 0.50 | 3.5 | 1.00 | 85 | 0.69 | 0.75 |
| 1 | bo0 | 13.0 | 0.84 | 8.0 | 0.21 | 88 | 0.41 | 0.44 |
| 2 | sobol | 18.6 | 0.78 | 6.3 | 0.52 | 80 | 0.26 | 0.22 |
| 3 | sobol | 17.4 | 0.72 | 6.2 | 0.82 | 70 | 0.59 | 0.59 |
| 4 | bo1 | 17.7 | 0.50 | 3.5 | 0.00 | 90 | 0.60 | 0.59 |
| 5 | sobol | 17.4 | 0.50 | 4.3 | 0.10 | 100 | 0.21 | 0.19 |
| 6 | sobol | 17.5 | 0.65 | 4.4 | 0.94 | 36 | 0.01 | 0.00 |
| 7 | sobol | 19.3 | 0.67 | 3.8 | 0.16 | 83 | 0.41 | 0.47 |
| 8 | rotfam | 17.5 | 0.70 | 6.0 | 0.50 | 112 | 0.51 | 0.56 |
| 9 | bo1 | 19.6 | 0.50 | 3.5 | 1.00 | 85 | 0.70 | 0.75 |
| 10 | sobol | 15.6 | 0.68 | 5.2 | 0.31 | 41 | 0.06 | 0.06 |
| 11 | bo0 | 13.0 | 0.84 | 8.0 | 0.00 | 90 | 0.39 | 0.44 |
| 12 | bo1 | 18.5 | 0.50 | 3.5 | 0.00 | 90 | 0.59 | 0.62 |
| 13 | sobol | 14.4 | 0.75 | 5.8 | 0.53 | 112 | 0.48 | 0.53 |
| 14 | mask:B45 | 15.0 | 0.70 | 6.0 | 0.75 | 45 | 0.03 | 0.03 |
| 15 | bo1 | 19.4 | 0.50 | 3.5 | 1.00 | 86 | 0.69 | 0.78 |

Table 12: The 16 designs of the portfolio (lengths in $\mu$m, $\rho$ = corner rounding, $\theta$ = rotation in degrees), with their yield on training and test worlds.

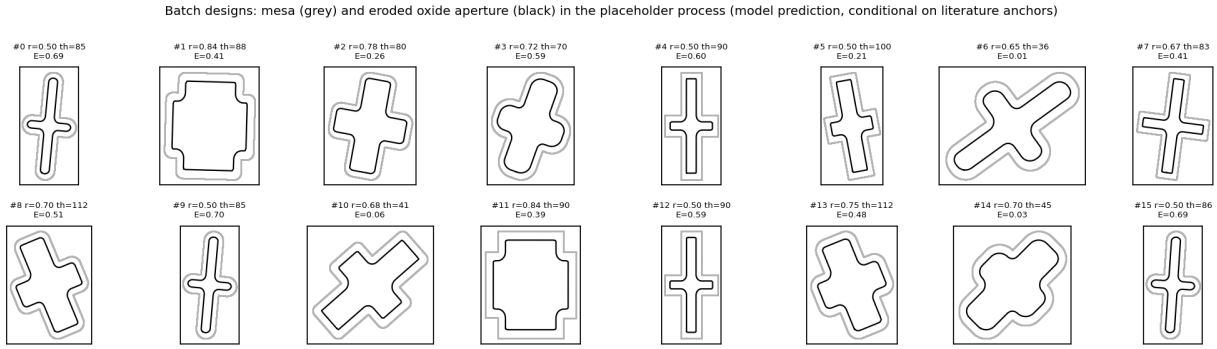



Figure 5: The portfolio: mesa (grey) and eroded oxide aperture (black) in the placeholder process.

**Boundary and fresh-device checks.** 8 of the 16 portfolio designs sit on a wall of the search box (thinnest arms and/or lowest arm ratio); their oxide apertures have an area of only 34–85 $\mu$m$^2$ against 79–88 $\mu$m$^2$ for the cruciforms of the experiments and 130 $\mu$m$^2$ for the circle (Table 14). Thin arms are where the scalar-mode, single-mode description is least credible and where loss and current density that the model does not contain would matter. Two checks were made on the same test worlds: (i) the sets were re-scored on a new population of 48 devices, so that the comparison cannot come from over-fitting to particular device draws; (ii) a portfolio was re-selected from the 81 candidates of the pool inside a conservative interior box (arm ratio $\geq 0.6$, $4.5 \leq w \leq 7.5\,\mu$m, $13 \leq L \leq 19\,\mu$m).

| set of designs (16) | $E$[best yield], fresh devices | $P$(some design $\geq 0.5$) |
|---|---|---|
| greedy portfolio, whole search box | 0.86 | 0.94 |
| greedy portfolio, interior box only | 0.81 | 0.91 |
| author's mask | 0.67 | 0.69 |
| seven designs of APL 126 | 0.58 | 0.59 |

Table 13: Fresh device population; same unseen worlds.

| # | $L$ | ratio | $w$ | $\rho$ | $\theta$ | aperture area ($\mu$m$^2$) | on a wall of the search box |
|---|---|---|---|---|---|---|---|
| 0 | 19.7 | 0.50 | 3.5 | 1.00 | 85 | 37 | yes |
| 1 | 13.0 | 0.84 | 8.0 | 0.21 | 88 | 84 | yes |
| 2 | 18.6 | 0.78 | 6.3 | 0.52 | 80 | 108 | |
| 3 | 17.4 | 0.72 | 6.2 | 0.82 | 70 | 88 | |
| 4 | 17.7 | 0.50 | 3.5 | 0.00 | 90 | 34 | yes |
| 5 | 17.4 | 0.50 | 4.3 | 0.10 | 100 | 48 | yes |
| 6 | 17.5 | 0.65 | 4.4 | 0.94 | 36 | 52 | |
| 7 | 19.3 | 0.67 | 3.8 | 0.16 | 83 | 49 | |
| 8 | 17.5 | 0.70 | 6.0 | 0.50 | 112 | 88 | |
| 9 | 19.6 | 0.50 | 3.5 | 1.00 | 85 | 37 | yes |
| 10 | 15.6 | 0.68 | 5.2 | 0.31 | 41 | 62 | |
| 11 | 13.0 | 0.84 | 8.0 | 0.00 | 90 | 85 | yes |
| 12 | 18.5 | 0.50 | 3.5 | 0.00 | 90 | 35 | yes |
| 13 | 14.4 | 0.75 | 5.8 | 0.53 | 112 | 67 | |
| 14 | 15.0 | 0.70 | 6.0 | 0.75 | 45 | 68 | |
| 15 | 19.4 | 0.50 | 3.5 | 1.00 | 86 | 36 | yes |

Table 14: Portfolio designs with the area of their oxide aperture in the placeholder process.

The advantage of the model-selected batch over the author's mask and over the seven published designs survives both checks (Table 13); the interior-only batch gives up part of it (0.83 on the original devices), which is the price of staying where the model is credible.

**What the batch tells the author about her mask.** Most of the value comes from a few designs: the first four designs of the greedy order already reach 0.81 on the training worlds and the first eight give nearly the same expected best yield on the test worlds as all sixteen (0.85 against 0.86; Table 9). Adding designs to the author's mask helps nearly as much as replacing it: Table 10 shows that two additions lift the expected best yield from 0.68 to about 0.85, close to the level of the whole portfolio, but the first of them is a thin-armed design on the wall of the search box, so it is a hypothesis rather than a recommendation. Restricting the additions to the interior box (Table 15) gives 0.78 with one added design, 0.80 with two and 0.83 with three, about the level of the interior-only sixteen-design portfolio (Table 13), and the first two additions are crosses close to mask family A (arm ratio 0.72–0.78, width about 6.2–6.3 $\mu$m) at 70° and 80°, rotations that the mask does not contain. The mask's rotations come in mirror pairs (30°/150°, 45°/135°, 60°/120°), which the model treats as equivalent up to device scatter: the pairs are replicates and a test of the mirror symmetry of the wafer (a difference between a design and its twin could point to strain, an off-cut substrate or misalignment), but they also mean that the mask samples only five distinct rotations.

| # | $L$ ($\mu$m) | arm ratio | $w$ ($\mu$m) | $\rho$ | rotation (°) | $E$(yield) alone | $E$[best] of mask + # |
|---|---|---|---|---|---|---|---|
| 1 | 17.4 | 0.72 | 6.2 | 0.82 | 70 | 0.59 | 0.78 |
| 2 | 18.6 | 0.78 | 6.3 | 0.52 | 80 | 0.26 | 0.80 |
| 3 | 18.4 | 0.65 | 5.1 | 0.80 | 91 | 0.57 | 0.83 |
| 4 | 17.5 | 0.70 | 6.0 | 0.50 | 112 | 0.51 | 0.83 |

Table 15: Additions to the author's mask restricted to the interior box, in the order of the greedy choice (chosen on the training worlds, scored on the test worlds).

**How optimistic is the headline?** The 32 test worlds were never used to choose the portfolio, but they are only 32 of the 181 accepted worlds. Scoring the same sets on the 117 accepted worlds that were used neither for selection nor for this test (same devices, same specification) gives lower levels and the same ordering (Table 16): portfolio 0.74, author's mask 0.61, seven published designs 0.56; portfolio minus mask +0.13 [+0.09, +0.16] (95 % bootstrap interval over worlds). The mask with three interior additions reaches 0.72 on these worlds (0.83 on the test worlds). The 0.86 is therefore an optimistic draw of 32 worlds; the level to quote is about 0.77 (all 149 worlds not used for selection), and it assumes that the best design of the batch can be identified after measurement.

| set of designs | 32 test worlds (headline) | 117 other accepted worlds | all 149 |
|---|---|---|---|
| greedy portfolio (16) | 0.86 | 0.74 | 0.77 |
| author's mask (16) | 0.68 | 0.61 | 0.63 |
| seven designs of APL 126 | 0.57 | 0.56 | 0.56 |
| rotation-only family (8) | 0.71 | 0.60 | 0.62 |
| author's mask + 3 interior additions | 0.83 | 0.72 | 0.74 |

Table 16: Expected best-of-batch yield on three sets of accepted worlds, none used to select the designs (`scripts/other_worlds_check.py`).

# 7 Sensitivity to the assumptions

Each row perturbs the model or re-conditions it on the literature, and re-scores three sets of designs: the portfolio, the author's mask and the seven designs of [10]. What matters is the *order* of the sets, not the absolute level. The portfolio was selected once, under the baseline assumptions. The first rows keep the 32 test worlds and change one assumption. The rows from "worlds accepted with LOOSE thresholds" on use 32 newly accepted worlds: under another translation of the anchors into thresholds (loose, strict; Sec. 3), under the wider prior of Sec. 3, or ("RECONDITIONED") after an assumption was changed *and* the four constants were re-inferred from the same literature outcomes (4000 prior draws each). The spin-flip constants of [2] accept no world (0 of 4000 draws), so for them only the row with the old worlds exists.

| scenario | portfolio (16) | author's mask (16) | APL 126 seven |
|---|---|---|---|
| baseline | 0.86 | 0.68 | 0.57 |
| process depth=0.7 | 0.85 | 0.71 | 0.64 |
| process depth=1.3 | 0.75 | 0.66 | 0.59 |
| process ox_ratio=1.0 | 0.75 | 0.82 | 0.77 |
| process ox_ratio=1.3 | 0.81 | 0.65 | 0.55 |
| dn_eff=0.005 | 0.82 | 0.67 | 0.54 |
| dn_eff=0.02 | 0.77 | 0.68 | 0.61 |
| chi_th=-0.2 | 0.19 | 0.27 | 0.24 |
| chi_th=-0.05 | 0.72 | 0.70 | 0.69 |
| chi_th=+0.05 | 0.29 | 0.18 | 0.13 |
| chi_th=+0.2 | 0.04 | 0.05 | 0.04 |
| coupling x0.5 | 0.25 | 0.43 | 0.67 |
| coupling x2 | 0.39 | 0.25 | 0.20 |
| device scatter x2 | 0.68 | 0.57 | 0.53 |
| mask tolerance (rot 1 deg, L and w 0.3 um) | 0.81 | 0.65 | 0.39 |
| fine geometry grid | 0.87 | 0.73 | 0.61 |
| window 1.15-1.60 (low bias) | 0.82 | 0.54 | 0.44 |
| window 1.15-2.05 (wide) | 0.89 | 0.76 | 0.67 |
| SFM constants of the metasurface paper (same worlds) | 0.00 | 0.00 | 0.00 |
| worlds accepted with LOOSE thresholds | 0.77 | 0.71 | 0.65 |
| worlds accepted with STRICT thresholds | 0.92 | 0.80 | 0.69 |
| worlds accepted under a wider prior (native dichroism from 0.2 1/ns) | 0.72 | 0.63 | 0.63 |
| RECONDITIONED: oxidation isotropic (ox_ratio=1.0) | 0.61 | 0.61 | 0.56 |
| RECONDITIONED: oxidation anisotropy ox_ratio=1.3 | 0.77 | 0.71 | 0.62 |
| RECONDITIONED: thermal dichroism drift chi_th=-0.2 | 0.17 | 0.24 | 0.11 |
| RECONDITIONED: thermal dichroism drift chi_th=+0.2 | 0.09 | 0.07 | 0.03 |

Table 17: Expected best-of-batch yield under perturbations (32 worlds per row).

**Reading of Table 17.** The ordering portfolio > mask > seven published designs holds in 19 of 26 rows (the baseline is one of the rows, and in the row with the spin-flip constants of [2] no set switches at all), but the rows are not equally informative; the absolute level is not stable at all (the portfolio's expected best yield runs from 0.00 to 0.92). *Robust (baseline and 14 rows).* Oxidation depth 0.7 and 1.3 $\mu$m, oxidation-rate ratio 1.3 (also when the constants are re-inferred), index step, device scatter, mask tolerance, finer geometry grid, the low-bias and the wide switching-current windows, loose and strict thresholds and the wider prior keep the portfolio ahead of the mask by 0.05–0.28 at an expected best yield of 0.68–0.92. *Ordering fails without re-inferring the constants (5 rows).* Isotropic oxidation ($\eta = 1.0$): mask 0.82, seven designs 0.77, portfolio 0.75. Half the coupling constants: seven 0.67, mask 0.43, portfolio 0.25. Thermal dichroism drift of $|\chi_{th}| = 0.2$: all three sets between 0.04 and 0.27. The spin-flip constants of [2] with the old worlds: no device switches in any set. These rows move a constant away from the values that reproduce the literature (halving the coupling, for instance, breaks the cancellation of

the native dichroism at arm ratio 0.7, the ridge of Fig. 1); the world is then no longer literature-consistent, so a low yield shows how tightly the design is tied to that ridge, not that the literature contradicts it. *Ordering holds only weakly or at low yields (3 rows).* $\chi_{th} = -0.05$ (0.72, 0.70, 0.69: indistinguishable), $\chi_{th} = +0.05$ (0.29, 0.18, 0.13), coupling $\times 2$ (0.39, 0.25, 0.20).

**Re-conditioned rows: what depends on the assumption itself.** *Oxidation.* With isotropic oxidation, 56 of 4000 draws still reproduce the literature outcomes, but the portfolio and the mask tie (0.61 each; seven designs 0.56): the advantage of a model-selected batch over the mask depends on the anisotropy of the oxidation. With $\eta = 1.3$ (70 accepted) it remains (0.77 against 0.71). The oxide anisotropy is measurable on a cross-section or an infrared image of the oxide front, far more cheaply than a mask run. *Thermal drift.* With $\chi_{th} = \pm 0.2$ (a dichroism that moves with current by an amount comparable to the native one across the operating range; the scale is an arbitrary probe, not an estimate) the literature outcomes are still reproduced (116 and 126 worlds accepted), but no set of designs exceeds an expected best yield of 0.24 and the ordering is lost or meaningless (0.17, 0.24, 0.11 for $-0.2$; 0.09, 0.07, 0.03 for $+0.2$). The sign of the native birefringence also follows the sign of $\chi_{th}$: $b_0 > 0$ in 97 % of the accepted worlds for $\chi_{th} = -0.2$ and in 4 % for $+0.2$, against 95 % in the primary set. The "$b_0 > 0$" of Sec. 3 is therefore a consequence of assuming no thermal drift, not something the literature fixes, and **every yield in this study presumes that the switching comes from the spin-flip dynamics at a fixed anisotropy**. If the switching of the fabricated devices is thermally driven, these predictions do not apply. *Wider prior.* With native dichroism allowed down to $0.2\,\mathrm{ns}^{-1}$ the portfolio still leads the mask, but by 0.09 (0.72 against 0.63, the seven designs 0.63).

**Does the best rotation depend on how the literature was translated into constraints?** The 1:0.7 cross ($L = 17.5$, $w = 6\,\mu$m, $\rho = 0.5$) was scored at eleven rotations on the worlds of every conditioning above (Table 18; at most 40 worlds per set, the constants of [2] accept no world).

| world set | worlds | 40° | 50° | 60° | 70° | 80° | 90° | 100° | 110° | 120° | 130° | 140° | best | within 90 % of best |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| primary (every third world) | 40 | 0.03 | 0.09 | 0.33 | 0.53 | 0.49 | 0.47 | 0.49 | 0.54 | 0.32 | 0.09 | 0.02 | 110° | 70–110° |
| loose thresholds | 40 | 0.02 | 0.04 | 0.33 | 0.48 | 0.47 | 0.43 | 0.47 | 0.49 | 0.31 | 0.04 | 0.02 | 110° | 70–110° |
| strict thresholds | 40 | 0.04 | 0.11 | 0.38 | 0.58 | 0.67 | 0.63 | 0.67 | 0.59 | 0.38 | 0.11 | 0.04 | 80° | 80–100° |
| wider prior | 40 | 0.07 | 0.14 | 0.29 | 0.44 | 0.49 | 0.49 | 0.49 | 0.44 | 0.29 | 0.14 | 0.07 | 80° | 80–110° |
| re-inferred, isotropic oxidation ($\eta = 1.0$) | 40 | 0.06 | 0.11 | 0.24 | 0.44 | 0.46 | 0.49 | 0.47 | 0.45 | 0.23 | 0.11 | 0.06 | 90° | 70–110° |
| re-inferred, $\eta = 1.3$ | 40 | 0.05 | 0.10 | 0.21 | 0.42 | 0.53 | 0.53 | 0.53 | 0.42 | 0.20 | 0.09 | 0.05 | 80° | 80–100° |
| re-inferred, thermal drift $\chi_{th} = -0.2$ | 40 | 0.02 | 0.02 | 0.02 | 0.01 | 0.03 | 0.05 | 0.03 | 0.02 | 0.02 | 0.02 | 0.02 | 90° | 90–90° |
| re-inferred, thermal drift $\chi_{th} = +0.2$ | 40 | 0.01 | 0.01 | 0.03 | 0.03 | 0.01 | 0.01 | 0.01 | 0.03 | 0.03 | 0.01 | 0.01 | 70° | 70–70° |
| worst case over the 6 working sets (share of best) | | 0.04 | 0.09 | 0.40 | 0.79 | 0.91 | 0.87 | 0.90 | 0.78 | 0.38 | 0.08 | 0.03 | 80° | |

Table 18: Mean hard yield of the 1:0.7 cross by rotation from the native axis, in each set of worlds; last row: for each rotation the lowest share of the set's own best rotation among the 6 sets in which some rotation works (best mean yield $\geq 0.2$). $\theta$ and $180° - \theta$ are mirror images; a "best" of 70° or 110° is the same statement.

In the 6 working sets the yield forms the same plateau between 70° and 110° (approximately mirror-symmetric about 90°), and it is only 40–68 % of the set's best at 60° and 38–65 % at 120°. Where the maximum lies inside the plateau does depend on the set: 70°/110° in the primary and loose sets, 80°–100° in the strict, wider-prior and $\eta = 1.3$ sets, 90° with isotropic oxidation. Measured against the best rotation of each set, 80° (and its mirror image, 100°) never falls below 0.91 of that set's best, 90° below 0.87 and 70°/110° below 0.79. **In this model 80° (equivalently 100°) is the most robust single rotation, narrowly**, and there is no reliable ranking inside the 70°–110° plateau: these are point estimates from 40 worlds per set. Resampling the worlds (2000 bootstrap resamples, yields averaged with the mirror-image rotation) gives 80° the highest worst case in 80 % of the resamples (90°: 7 %, 70°: 13 %), 80° above 90° in 93 % and above 70° in 87 %; the worst-case share is 0.86 (95 % interval 0.70–0.97) at 80°, 0.81 (0.64–0.93) at 90° and 0.75 (0.61–0.88) at 70°. The preference for 80° over 90° is therefore indicative, not established; the firm statement is that 80°–100° is the core of the plateau, and the statement of Sec. 4 that 70°–80° is best is a statement about the primary set. With thermal drift (last two sets) no rotation works (best mean yield 0.05 or lower), so no rotation can be recommended if switching is thermally driven.

**What to measure first.** The rows above point at the same three unknowns, in the order of their leverage on the design: (1) the anisotropy and depth of the oxidation (cross-sections, infrared images of the oxide front); (2) whether the switching of a fabricated device follows the current or the temperature (polarization-resolved $L$–$I$

curves at several heat-sink temperatures); (3) the scale of the birefringence and dichroism that a given shape produces (polarization-resolved spectra below threshold, as in [4]).

### 7.1 Is the birefringence channel optical or mechanical? A semi-vectorial check

The model routes the shape dependence of the birefringence through a stress-weighted boundary tensor with a free coefficient $k_b$ (Sec. 2). One part of the birefringence can instead be computed without free coefficients: the *form* birefringence of the lateral index guide that the aperture defines. A semi-vectorial finite-difference solver (`src/vcsel_bo/vector_modes.py`) solves the transverse equation for each linear polarization, with the normal component $\varepsilon E_\perp$ continuous across index steps, on the same grid for both polarizations; solving on grids aligned with 0° and 45° gives the magnitude and axis of the splitting, converted as $\Delta\nu = \nu\,\Delta n_{\mathrm{eff}}/n_g$ and $\gamma_p = \pi\Delta\nu$ ($n_g = 3.6$ and index step $\Delta n_{\mathrm{eff}} = 0.01$ assumed). *Checks.* The solver reproduces the exact TE–TM splitting of a dielectric slab to 0.1 % (second-order convergence), gives no splitting for a circle, follows the long axis of an ellipse under rotation, and changes by 1.3 % between grids of 0.1 and 0.07 $\mu$m.

| Design | $\Delta n_{\mathrm{eff}}$ ($10^{-6}$) | axis (°) | split. (MHz) | $\gamma_p^{\mathrm{form}}$ (ns$^{-1}$) | $\lvert k_b \rvert \lvert \mathbf{s}_{stress} \rvert$ (ns$^{-1}$) [5–95 %] |
|---|---|---|---|---|---|
| A0 | 0.26 | 0.2 | 23 | 0.07 | 6.7 [1.2–32.9] |
| A60 | 0.33 | 61.6 | 29 | 0.09 | 6.8 [1.2–33.2] |
| A90 | 0.36 | 89.9 | 31 | 0.10 | 6.9 [1.2–33.5] |
| B0 | 0.51 | 0.1 | 46 | 0.14 | 9.8 [1.8–48.0] |
| B60 | 0.52 | 59.5 | 46 | 0.15 | 9.3 [1.7–45.5] |
| B90 | 0.53 | 89.9 | 47 | 0.15 | 9.1 [1.7–44.6] |

Table 19: Form birefringence of six designs of the author's mask (placeholder process: depth 1 $\mu$m, oxidation-rate ratio 1.15; all 16 designs and the isotropic process are in `results/form_birefringence.json`), compared with the phenomenological stress term over the literature-consistent worlds. MODEL PREDICTIONS.

**Reading.** Over all 16 designs the form birefringence is 0.07–0.15 ns$^{-1}$ (splittings of 20–50 MHz); with a five times larger index step (0.05) it stays at or below 0.4 ns$^{-1}$. Measured VCSEL splittings are 1.3–9.9 GHz ($\gamma_p \approx$ 4–31 ns$^{-1}$, [4]), and the stress term needed to reproduce the published outcomes is of the same order. The shape therefore acts on the birefringence mainly through strain, not through the optics of the guide, which supports routing it through a stress channel; but the $1/(r^2+1)$ kernel of that channel remains an assumption. The next simulation of value is a mechanical one (oxide shrinkage as the load, then the photoelastic tensor of GaAs), not a more accurate optical solver. The sign with which the form term would enter the SFM depends on the field-envelope convention and was not verified; the term is too small to change the results either way.

## 8 Testable predictions for the mask

The mask has two aperture families (A: 17.5 × 12.25 $\mu$m, corners assumed sharp; B: 15.0 × 10.5 $\mu$m, short-arm radius 2.25 $\mu$m; arm width 6 $\mu$m) at eight rotations, 32 devices in all, i.e. two per design if all designs are equally represented. Table 20 lists what the 181 literature-consistent worlds predict; the median and the 10–90 % interval over worlds are given, because the constants are not known.

| design | yield: median [10%, 90%] | lases first on $x$ (median share) | azimuth: axial mean ($R$) | switching share (median) | median $\mu_{sw}$ |
|---|---|---|---|---|---|
| A, 0° | 0.00 [0.00, 0.00] | 1.00 | 180° (1.00) | 0.00 | 1.90 |
| A, 30° | 0.00 [0.00, 0.00] | 1.00 | 3° (0.76) | 0.00 | 1.45 |
| A, 45° | 0.00 [0.00, 0.27] | 1.00 | 6° (0.60) | 0.00 | 1.28 |
| A, 60° | 0.22 [0.00, 0.90] | 0.19 | 75° (0.45) | 0.60 | 1.28 |
| A, 90° | 0.28 [0.03, 0.94] | 0.00 | 90° (1.00) | 0.70 | 1.69 |
| A, 120° | 0.23 [0.00, 0.84] | 0.22 | 109° (0.47) | 0.58 | 1.29 |
| A, 135° | 0.00 [0.00, 0.23] | 0.99 | 173° (0.60) | 0.01 | 1.24 |
| A, 150° | 0.00 [0.00, 0.00] | 1.00 | 176° (0.76) | 0.00 | 1.44 |
| B, 0° | 0.00 [0.00, 0.00] | 1.00 | 180° (1.00) | 0.00 | 1.75 |
| B, 30° | 0.00 [0.00, 0.00] | 1.00 | 9° (0.73) | 0.00 | 1.40 |
| B, 45° | 0.00 [0.00, 0.14] | 0.94 | 20° (0.49) | 0.00 | 1.31 |
| B, 60° | 0.03 [0.00, 0.86] | 0.00 | 78° (0.82) | 0.35 | 1.43 |
| B, 90° | 0.00 [0.00, 0.76] | 0.00 | 90° (1.00) | 0.00 | 1.98 |
| B, 120° | 0.04 [0.00, 0.78] | 0.01 | 103° (0.81) | 0.31 | 1.46 |
| B, 135° | 0.00 [0.00, 0.17] | 0.88 | 158° (0.51) | 0.00 | 1.29 |
| B, 150° | 0.00 [0.00, 0.00] | 1.00 | 171° (0.74) | 0.00 | 1.31 |

Table 20: Predictions for the 16 designs of the mask (rotation from the native axis $x$). Azimuth of the low-current state in the crystal frame: axial mean over worlds (0° and 180° are the same direction) with the resultant length $R$ in brackets ($R$ near 1: the worlds agree; near 0: they do not).

| comparison (yield) | $P$(first larger) | $P$(within 0.1) |
|---|---|---|
| A90 > A60 | 0.61 | 0.29 |
| A90 > A45 | 0.87 | 0.25 |
| A60 > A45 | 0.73 | 0.44 |
| A90 > A30 | 0.94 | 0.24 |
| A60 > A120 | 0.38 | 0.97 |
| A45 > A135 | 0.08 | 1.00 |
| A30 > A150 | 0.05 | 1.00 |
| A90 > B90 | 0.87 | 0.25 |
| A60 > B60 | 0.46 | 0.31 |
| A45 > B45 | 0.15 | 0.81 |

Table 21: Pairwise orderings of yield across worlds.

**Predicted numbers of switching devices (two devices per design).** Drawing a world and then two Bernoulli devices per design, the model predicts for the 12 devices of the 60°, 90°, 120° designs a mean of 5.5 switching devices (5th–95th percentile: 1–11), for the 12 devices at 0°, 30°, 150° a mean of 0.5 (0–4), and for the 8 devices at 45°, 135° a mean of 1.3 (0–5). If the arm tips of family A were fully rounded ($\rho = 1$) instead of sharp, the means become 5.0 (0–11) at 60°–120°, 0.4 near 0° and 1.2 at 45°, 135° (`scripts/mask_family_a_rounded.py`): the corner treatment shifts the numbers a little but does not change the picture. Here a device *switches* if it shows at least one polarization flip in the sweep; the number of devices that also meet the whole yield specification (one switch inside the current window, stable states, separation, ellipticity) is smaller: 3.4 of the 12 devices at 60°–120° (0–8) and 0.2 of the 12 near 0°. **How the predictions can fail.** A measurement outside these intervals — more than 4 of the 12 devices near 0° switching, or none of the 12 devices at 60°–120° — lies beyond the 5th or 95th percentile of the prediction, so it would count against the model class at about the 5 % level for each statement; it would not by itself refute it, because the placeholder oxidation, the two devices per design and the assumed sharp corners of family A could equally be at fault. The near-0° group contains the 30° and 150° designs, and switching of the 30° cross is a published observation that the model does not reproduce (Sec. 3), so a switching 30° device would not be a new failure. The intervals are wide because the literature-consistent set is wide; with the first measurement they are expected to narrow, not to close. Design-level statements are weaker still: with two devices, one design cannot distinguish a yield of 0.3 from 0.7. Measuring at least eight devices for the four designs 60° and 90° of family A and B would be a natural extension (the author's judgement, not a computed optimum). Further qualitative predictions: in the median world the low-current polarization follows the long arm ($x$ for 0°–30°, $y$ for 60°–120°), but family A at 60° and 120° lases first on $x$ in 28 % and 29 % of the worlds (family B: 8 % and 9 %), so this is a probabilistic prediction; mirror-image rotations ($\theta$ and $180° - \theta$)

behave alike within scatter (Table 21: e.g. 30° against 150°).

**Calibration protocol.** For every fabricated design record, from polarization-resolved $L$–$I$ sweeps (1.05–2.1 $I_{th}$, fixed temperature): the number of devices $n$, the number with exactly one polarization switch, the median switching current $\mu_{sw}$ of the switchers, and the azimuth of the low-current state in the crystal frame ($x$ = wafer flat). Then `vcsel_bo.calibration.abc_posterior(observations, worlds=...)` re-weights the literature-consistent worlds by how well their simulated summaries reproduce the measured ones (Gaussian-kernel ABC [20]); the batch design of Sec. 6 is re-run on the posterior worlds. The module is exercised on synthetic data only (`tests/test_calibration.py`).

**What could a first batch teach? (synthetic test, well specified, therefore optimistic).** The wafer is assumed to be one of 64 literature-consistent worlds; the mask is "measured" with $n$ devices per design; the other worlds are re-weighted by the likelihood of the counts (and, in a second model, of the azimuth and switching current); the pool design with the largest posterior expected yield is chosen and scored in the true world. Without any measurement the best single design of the pool has a mean yield of 0.64; the oracle that knows the true world reaches 0.86.

| observation model | tempering $T$ | devices per design | yield of the chosen design | effective sample size | $P$(true sign of $k_b$) |
|---|---|---|---|---|---|
| counts | 1 | 2 | 0.70 | 3.0 | 0.82 |
| counts | 1 | 4 | 0.69 | 2.0 | 0.84 |
| counts | 1 | 8 | 0.68 | 1.5 | 0.87 |
| counts+shape | 1 | 2 | 0.68 | 1.7 | 0.88 |
| counts+shape | 1 | 4 | 0.68 | 1.4 | 0.91 |
| counts+shape | 1 | 8 | 0.69 | 1.3 | 0.91 |
| counts+shape | 4 | 2 | 0.69 | 3.5 | 0.85 |
| counts+shape | 4 | 4 | 0.70 | 2.8 | 0.86 |
| counts+shape | 4 | 8 | 0.70 | 2.2 | 0.88 |

Table 22: A synthetic first batch. Yield of the design chosen after the measurement (no measurement: 0.64; oracle: 0.86). Prior probability of the correct sign of $k_b$ from the base rate: 0.83.

The gain is modest — from 0.64 to 0.68–0.70 — and does not grow with more devices per design, because the posterior collapses onto very few of the 64 worlds (effective sample size 1.3–3.5); tempering the likelihood does not change this. The sign of the birefringence coupling $k_b$, which the literature leaves open, is recovered with a probability of at most 0.91, against 0.83 for always guessing the more frequent sign. Two lessons: a continuous calibration model rather than a finite set of worlds is needed once measurements exist, and the azimuth of the low-current state and the switching current should be recorded, not only whether a device switches.

# 9 Limitations and what remains uncertain

- Constants and process parameters are not measured; the anchors are qualitative and many worlds satisfy them. Conclusions are therefore stated as distributions over worlds, never as one fitted value.
- Thresholds that translate "most/seldom/part" into numbers are choices (Sec. 7 shows their effect); the sizes of the fabricated apertures are not in the text read, so the published designs are drawn at assumed sizes ($L = 17.5\,\mu$m, $w = 6\,\mu$m, corner fraction 0.5).
- Structural limits: single-mode SFM although apertures of this size may be multimode; scalar mode used only as a weight; linear phenomenological coupling; second-rank symmetry; the birefringence channel is attributed to stress, which a semi-vectorial check supports only indirectly (the optical form birefringence is 25–450 times smaller than measured splittings at the assumed index step, and at least 10 times smaller at a five times larger step, Sec. 7.1) and which no mechanical simulation has computed; two of six published observations not reproduced (Sec. 3).
- Sources not read in full: [6] (abstract only), [3] (listing only). The SFM constants of [10] could not be re-checked in the last pass; the model uses typical magnitudes. The second set, from [2] (spin-flip rate 20 times larger), accepts no world in 4000 draws, so that robustness check is inconclusive, not confirmatory.

- The corner treatment of mask family A is not recorded in the project notes and is assumed sharp; family B has the recorded 2.25 $\mu$m short-arm radius. With fully rounded arm tips for family A the predicted number of switching devices at 60°–120° falls from 5.5 to 5.0 of 12 (Sec. 8); the author should check the mask file.
- The yields presume that switching comes from the spin-flip dynamics at a fixed anisotropy; a thermal dichroism drift of the size probed in Sec. 7 removes the yield from every set of designs, and the oxidation anisotropy (placeholder 1.15) decides whether the model-selected batch beats the author's mask.
- The accepted native dichroism piles up against the lower edge of the default prior box (Sec. 3); the wider prior changes the numbers but not the ordering of the design sets, and in the re-conditioned worlds the sign of the native birefringence depends on the assumed thermal drift.
- The headline batch scores are optimistic. The best-of-batch yield is 0.86 on the 32 test worlds but 0.74 on 117 further accepted worlds (Sec. 6); the ordering of the sets holds, the level does not, and "best of batch" assumes that the best design can be identified after measurement.
- The optimiser benchmark has five simulated wafers for P1 and P2 (chosen among feasible accepted worlds to span the range of difficulty) and eight runs for P3; runs on one wafer are not independent, so the paired-run tests overstate the evidence (Sec. 5).
- No claim about real devices is made; nothing has been validated by measurement.

# 10 Reproducibility

`pip install -r requirements.txt`, `pytest` (unit, regression and citation-consistency tests); `scripts/run_all.sh` regenerates the main results (roughly 1.5 hours on two cores; the logged steps took about 75 minutes) and `scripts/run_extra.sh` the robustness extras of Secs. 6–7 (about 20 minutes); `scripts/form_birefringence.py` the semi-vectorial check of Sec. 7.1 (about 1 minute); `scripts/make_figures.py`, `scripts/make_tables.py` and `scripts/make_readme.py` rebuild the figures, the tables of this document and the README. The primary set of literature-consistent worlds took 448 s on two cores. All random streams are seeded.

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